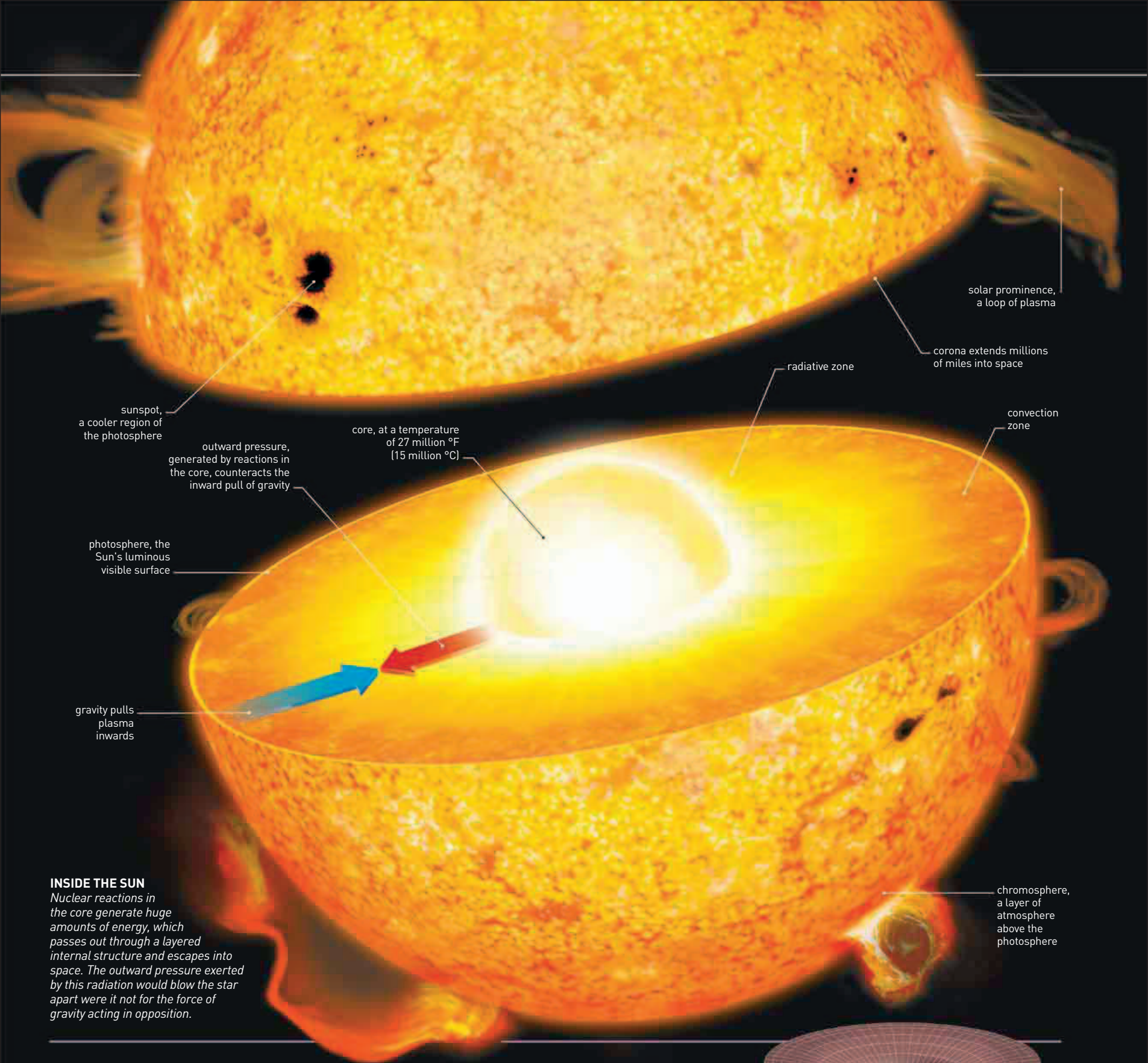




INSIDE THE SUN

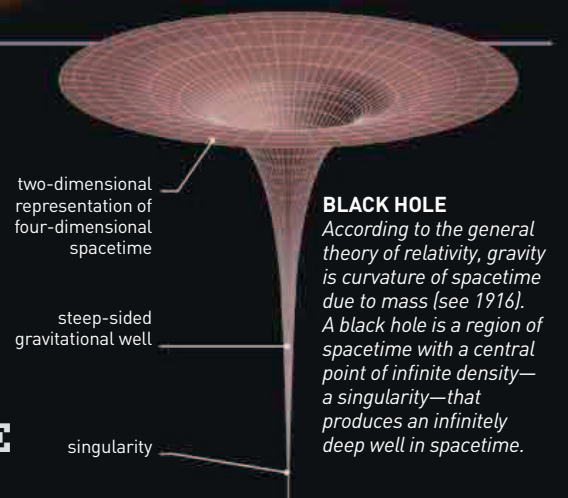
Nuclear reactions in the core generate huge amounts of energy, which passes out through a layered internal structure and escapes into space. The outward pressure exerted by this radiation would blow the star apart were it not for the force of gravity acting in opposition.

NEUTRON STARS AND BLACK HOLES

Toward the end of a star's life, nuclear fusion falters. The star starts to cool and collapse under its own gravity. Inside a star like the Sun, a force called electron degeneracy pressure resists further collapse and the star becomes a white dwarf. However, in some more massive stars, gravitational collapse overcomes this force and pushes electrons and protons together to form neutrons. The result is a neutron star, which is prevented from further collapse by a force called neutron degeneracy pressure. In a very massive

star, even this force cannot halt collapse and the star continues to shrink, eventually becoming a black hole—a region of spacetime so dense that even light cannot escape from it.

160,000
THE NUMBER OF LIGHT-YEARS FROM EARTH TO THE NEAREST BLACK HOLE



BLACK HOLE

According to the general theory of relativity, gravity is curvature of spacetime due to mass (see 1916). A black hole is a region of spacetime with a central point of infinite density—a singularity—that produces an infinitely deep well in spacetime.